

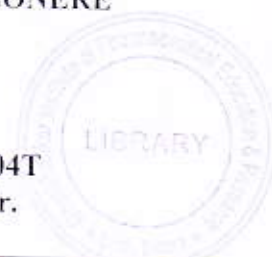
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2024

Date: 26-12-2024

Course: B. Pharmacy
Subject Name : Novel Drug Delivery System
Max Marks : 75

Sem: VII
Subject Code: BP704T
Duration : 3 Hr.



Instructions:

1. All the questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

- Q. 1. Objective type questions (Answer all the questions)** (2 x 10) = 20
- i) Classify approaches for design of controlled drug delivery system.
 - ii) Give the advantages and limitations of microencapsulation.
 - iii) Define mucoadhesion and enlist theories of mucoadhesion.
 - iv) What are the properties of a drug that make it suitable for the preparation of a transdermal drug delivery system?
 - v) Classify the various approaches for gastroretention.
 - vi) Define implants and give its different types.
 - vii) Give the advantages and limitations of IUDs.
 - viii) Outline the classification of polymers.
 - ix) Define permeation enhancers and give examples.
 - x) Enlist Excipients used in Nasal spray formulations.
- Q. 2. Long answer questions (Answer any 2 out of 3)** (10 x 2) = 20
- i) Define targeted drug delivery system and give its advantages and limitations. Write a detailed note on nanoparticles as targeted drug delivery system.
 - ii) Define microencapsulation and give its advantages and limitations. Explain coacervation-phase separation technique.
 - iii) What is gastroretentive drug delivery system, give its advantages and limitations and describe non-floating systems.
- Q. 3. Short answer questions (Answer any 7 out of 9)** (5 x 7) = 35
- i) Explain different formulation approaches of transdermal drug delivery system.
 - ii) Describe dry powder inhalers (DPI) in detail.
 - iii) Explain various barriers for ocular drug delivery system.
 - iv) Write a note on intrauterine devices.
 - v) Explain physiological aspects in buccal drug delivery system.
 - vi) Describe passive implants in detail.
 - vii) Explain the different theories of Mucoadhesion.
 - viii) Enlist Physicochemical and biological properties of drugs relevant to controlled release formulations and explain biological properties.
 - ix) Describe monoclonal antibodies and their applications.

-----END OF THE PAPER-----